

Project Design Phase-II

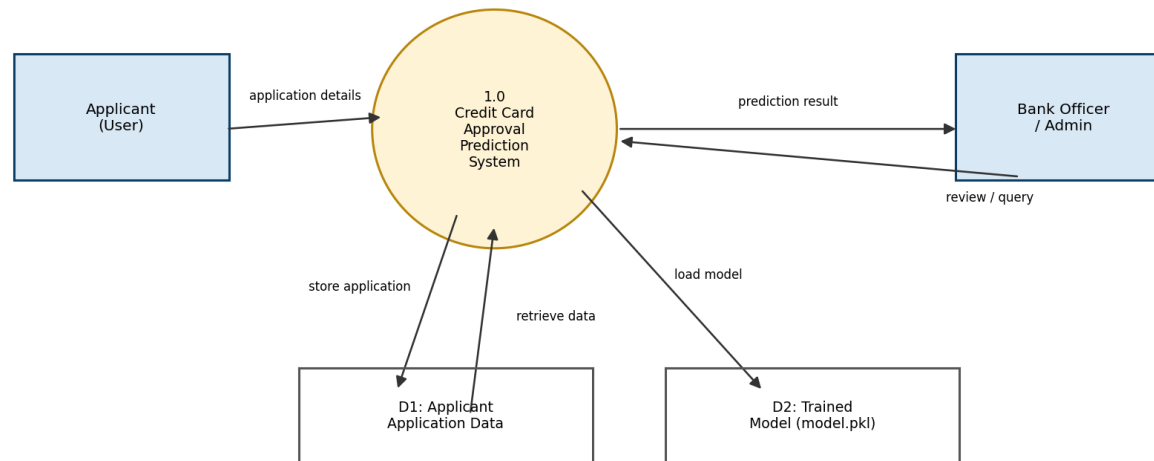
Data Flow Diagram & User Stories

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Data Flow Diagram:

The diagram below shows how applicant data flows through the Credit Card Approval Prediction System - from the applicant, through the prediction process, to the stored application data and trained model, and finally to the bank officer / admin who reviews outcomes.

DFD Level 0: Credit Card Approval Prediction System



User Stories:

Use the below template to list all the user stories for the product.

User Type	Functional Req. (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Applicant (Web user)	Application Submission	USN-1	As an applicant, I want to fill in my personal and financial information through a web form so I can apply for a credit card.	The form accepts correctly filled data and displays an error when required fields are missing or invalid.	High	Sprint-1
Applicant (Web user)	Application Submission	USN-2	As an applicant, I want to submit my finished application form to get an approval prediction.	Submitting the form starts the prediction pipeline and displays the results page.	High	Sprint-1
Applicant (Web user)	Prediction Result	USN-3	As an applicant, I want to instantly see if my application has been Approved or Rejected right after I submit it.	The results page clearly shows the Approved or Rejected outcome.	High	Sprint-1
Bank Officer (Admin)	Model Training	USN-4	As a bank officer, I want the system trained using past applicant records so predictions match real-world approval trends.	The model gets trained on the dataset, and the evaluation metrics are recorded.	High	Sprint-2
Bank Officer (Admin)	Model Evaluation	USN-5	As a bank officer, I want to know which machine learning model gives the best results so I can rely on the deployed predictions.	Documentation includes a table comparing Accuracy, Precision, Recall, and F1-Score for each model.	Medium	Sprint-2
Bank Officer (Admin)	Error Handling	USN-6	As a bank officer, I want the system to manage incorrect inputs smoothly rather than failing unexpectedly.	Missing or incorrect form fields trigger a clear error message rather than a server failure.	Medium	Sprint-2